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Project Milestone 1 - Decision and System Definition

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1. Project Title and Context

Working Title: Heat-Risk Activation: Decision-Support System for Municipal Heat-Response Interventions.

Extreme heat is the leading cause of death in the United States and around the world, and its public-health burden shows up far more often as emergency department (ED) visits and hospitalizations than as mortality. There are approximately 67,000 emergency room heat-related ED visits [1], 9200 hospitalizations[2], and 1700 deaths across the US on average each year[3]. City public health and emergency management offices already operate cooling centers and issue heat advisories to their communities, but the decision on when and where to activate these existing resources ahead of forecasted heat events is currently still made with limited, informal use of forecasting data.

This project studies the City of Phoenix, Arizona as the city of analysis, and addresses how forecast, historical illnesses, and community-vulnerability data can support a more systematic activation decision. Phoenix is chosen for this study as it is considered the hottest major city in the US[4]. County-day data (Maricopa County, day t) will be used as the unit of study, where outcome and vulnerability data are published. This is because data from the CDC report data at county granularity, not city granularity. The boundary of this semester's project is the activation decision for existing cooling infrastructure. A previous drafted scope of this project was to decide whether to build new cooling centers for cities. However, the time frame to build new cooling centers is incompatible with a single heat-wave forecast's lead time, therefore, the project decision has been modified.

2. Decision or Scientific Question

[1] https://www.climatecentral.org/climate-matters/heat-illness-emergency-room-visits

[2] https://www.sciencedirect.com/science/article/pii/S2667278223000561#bib0002

[3] https://www.npr.org/2026/08/31/nx-s1-5724225/heat-wave-death-toll-tracker

[4] https://www.accuweather.com/en/weather-news/10-of-the-hottest-cities-in-the-us/432421

Central decision: Should the city activate a heat-response intervention for the City of Phoenix, Arizona ahead of a forecasted heat event beginning at time *t*? Heat-response interventions include staffing and opening existing cooling centers, and issuing targeting outreach to increase awareness on heat prevention.

What must be decided: which action $a \in A$ to take for zone z, given a heat-wave forecast issued at time t. Actions $a \in A$ includes no action, public advisory, activate cooling centers, or targeted outreach to high-vulnerability zones.

When: The question must be answered at forecast issuance, 3-7 days before a forecasted excessive-heat event.

Unit: county-day (zone *z,* day *t*)

What would constitute a useful answer: An activation decision for each forecasted zone, that measurably reduces heat-related ED visits relative to a no-action baseline. The decision also has to be made on resource costs that the city can sustain.

3. Stakeholder, Decision Maker, and Intended Value

Primary stakeholder: City of Phoenix Office of Heat Response and Mitigation, Phoenix Office of Emergency Management. These stakeholders are currently active, and responsible in decision-making that publishes annual Heat Response Plans

Who will use this information: City of Phoenix Office of Heat Response and Mitigation staff responsible for escalating heat-response protocols

What are they trying to accomplish: Reducing heat-related illness burden on the health system. This is subject to limited staffing/ operating budget for cooling centers

How analytical information could improve the current process: Temperature-only trigger with a forecast, and vulnerability-informed estimate of heat-related ED-visit risk.

Value created: fewer avoidable heat-related ED visits or hospitalizations, and more efficient allocation of limited staffed cooling-center hours

What the system is not intended to do: decide where to build new cooling centers, replace heat-related clinical treatments

4. Alternative Actions and Expected Outcomes

There are feasible actions for each zone forecast. The following combinations of a1-a3 are possible, where a1 is the comparison baseline:

- $a0$ – no additional action taken
- $a1$ – issue public heat advisory/ awareness messaging
- $a2$ – activate and staff existing cooling centers in zone z
- $a3$ – target outreach to identified zones with high-vulnerability populations

Outcomes or consequences associated with these actions:

- High predicted heat risk + actions (a2/a3): Expecting reductions in ED visits. This would cost staffing and operating cooling center expenses
- High predicted heat risk + no action: Same or increase in ED-visit/ hospitalization risks, especially among identified high-vulnerability populations
- Overestimated risk + action: Staffing and operational costs wasted
- Forecast system not available: Revisit and carry out city's existing standard heat-response protocols

5. Information Available at Decision Time

Available before decision: CDC Heat & Health Tracker for Maricopa County-level heat-wave forecast[5], historical heat-related ED-visit and hospitalization rates for Maricopa County, CDC Vulnerability Index[6] for Phoenix area, existing cooling centers, location and capacity (from City of Phoenix Heat Response Plan[7]

Generated by analytical workflow: Predicted probability that zone z exceeds a defined heat-related ED-visit rate threshold within the forecast horizon.

Available only after the decision or outcome: Realized ED-visits and hospitalization counts for the forecasted period, whether actions taken (opening and staffing cooling centers and community awareness) were used at capacity.

Desirable but not actually available: Real-time cooling center occupancy, causal-effect estimate of activation actions

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[5] https://ephtracking.cdc.gov/Applications/heatTracker/

[6] https://www.atsdr.cdc.gov/place-health/php/svi/index.html

[7] https://www.phoenix.gov/administration/departments/heat/heat-response-programs/heat-response-plan.html

6. Decision-System Canvas v2

Component (Known, Hypothesis, Unresolved)	Layman Explanation	Mathematical/ Technical Explanation
1. Decision Context (Known)	The City of Phoenix Office of Heat Response (OHRM), with the Office of Emergency Management, decides whether to escalate heat-response intervention for zone z (Phoenix sub-area) ahead of forecasted heat event at time t. Objective: reduce heat-related ED-visits and hospitalization at sustainable operating costs.	Decision maker selects $a \in A$, given heat-forecasted context at forecasted issuance time t.
2. Data-Generating System (Hypothesis)	Environmental heat conditions combined with population exposure and community vulnerability to produce heat-related illness outcomes (ED-visits, hospitalizations)	$H_{\{z,t\}}\,(heat\ index) \rightarrow E_{\{z,t\}} = f(H_{\{z,t\}}, V_z)\,(exposure) \rightarrow Y_{\{z,t\}} = g(E_{\{z,t\}}, V_z)\,(outcome)$, where V_z is community vulnerability
3. Observations (Known)	Forecasted heat index for Phoenix, historical ED-visits/ hospitalization rates and Social Vulnerability Index (SVI) at Maricopa County, existing cooling-center capacity per published Phoenix Heat Response Plan	$X_{\{z,t\}} = \{forecast\ heat\ index_{\{z,t\}}, history\{Y_{\{county,\,s\}}: s < t, V_{\{county\}}, existing\ capacityy_z$
4. Learning (Hypothesis)	Learn how probability of elevated heat-related ED-visits and hospitalization rate depends on forecasted heat conditions and community vulnerability	
5. Prediction/ Inference (Hypothesis)	This is a prediction problem, estimating the probability that zone z exceeds the ED-visit-rate threshold within the horizon	
6. Validation/ Uncertainty (Unresolved)	Factors that create uncertainty: Age, baseline health, housing/ cooling access, reporting completeness. Testing generalization: time-based holdout	

	across heat seasons	
7. Alternative Actions (Known)	For each zone heat forecast, the city can choose among a list of defined actions of real interventions using existing resources	A = {a0 = no action, a1 = public advisory, a2 = activate/staff existing cooling centers, a3 = targeted outreach to high-SVI sub-populations}.
8. Decision (Hypothesis)	If predicted risk is high enough, escalate to a stronger action. The threshold has to trade off the cost of unnecessary activation against cost of unaddressed elevated-risk events	$a^*_{\{z,t\}} = a2/a3$ if $\hat{p}_{\{z,t\}} \geq n, \; else \; a1/\; a0$, where n is chosen by expected-cost minimization
9. Action (Known)	Emergency-management office executes the selected action from above	Execution of $a^*_{\{z,t\}} \in A$ at time t
10. Observed Outcome (Known)	Observe the realized heat-related ED-visit rate (primary outcome) and hospitalization (secondary outcome) for zone z	$Y^{obs}_{z,t+\Delta}$ (ED-visit rate), $Y'^{obs}_{z,t+\Delta}$ $(hospitalization\, rate)$
11. Monitoring/ Learning/ Adaptation (Unresolved)	After each heat season, check weather predicted risk matches realized outcomes. Check whether any subgroup was systematically under-predicted, and whether n is still appropriate	

7. Initial Assumptions

- Maricopa County-level ED-visit rates and vulnerability indicators are a reasonable way to determine Phoenix-specific risk. As Phoenix accounts for roughly a third to two-fifths of the county's population. This is an assumption and should be confirmed later.
- Relationship between forecasted heat index, community vulnerability index, and heat-related ED-visit rate is stable enough over historical periods to be predictive moving forward.
- CDC Heat & Health Tracker ED-visit data is timely and complete enough for decision time use.
- Existing cooling-center capacity is at least sufficient to support individuals in the highest-risk population when cooling centers are activated.

- Forecasted heat information is available with enough time window (3-7 days) to execute staffing decision before the event

8. Project Work Plan and Milestone

Timeline starts after Milestone 1 is due:

- Week 1-2: Data Acquisition: Collect CDC Heat & Health Tracker historical county-level ED-visit/ hospitalization rates, and heat-wave forecast archives
- Week 3-4: Data understanding and data quality assessment. Checking data completeness, missing patterns and alignment of forecast with realized heat index
- Week 5-6: Exploratory analysis. Examining relationship between forecasted heat index, SVI and historical ED-visit rate.
- Week 7-8: Baseline and prediction model implementation
- Week 9: Decision integration. Translating predicted risk into activation decision using cost trade-off to select a threshold n.
- Week 10: Uncertainty analysis
- Week 11: Reproducibility and System Integration
- Week 12: Final Technical Defense. Prepare and rehearse semester defence, emphasize unresolved items flagged in each milestone.

9. Major Technical Risk

1. City/ County granularity mismatch: The decision-maker city (Phoenix) and action set (Phoenix cooling centers) operate at the city level, but the outcome and vulnerability data are published at the Maricopa County level. This is a gap between what is being studied, and where the data is being pulled from. This matters because the gap can cause shifts in the information that is being analyzed. To evaluate this, I will compare Phoenix's population share the demographic profile against the countywide figures, and look for any Phoenix-specific heat-illness surveillance that could substitute or validate the county-level information. Alternatives would include Phoenix-specific outcome series data.
2. Confounding between forecast skill and true exposure: The forecast heat index is already a prediction, and not the realized exposure. A risk here would be systematic bias on

forecasting. To detect or evaluate this, I would have to compare the forecast vs a realized heat index from historical data at lead times used for decisions. An alternative plan is to model forecast realization errors more explicitly as an uncertainty component rather than treating the forecast as it is.

10. AI Use Statement

Generative AI (Claude) was used during the development of Milestone 1 submission. Claude was used to help structure the Decision-System Canvas v2 against the eleven-component course architecture. Generative AI was also used to further analyze instructor feedback on an earlier version of this project idea and to implement it into this current version. Generative AI was also used to structure this Milestone 1 dossier. All analytical framing, scope decisions and assumptions were all reviewed and checked by the student.